

Transmitter

Information
bits

1 0
1 1

(8,8)-16APSK
modulation

s_k

Atmospheric FSO channel

Block fading



Carrier phase

CFO · Doppler · PN

Complex
AWGN



amplitude factor -

phase

additive

Composite FSO channel

r_k

Channel blocks

100 symbols

DSP window

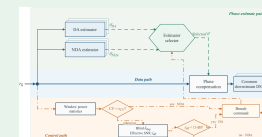
256 samples

Within one DSP window



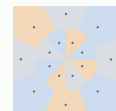
Coherent receiver DSP

Adaptive carrier
recovery



Detailed in Fig. 2

16APSK
demodulation



Detected
bits